STEP 1 - CREATE NOTEBOOK

STEP 2 — Install Required Libraries

```
!pip install spacy transformers torch pandas datasets
!python -m spacy download en_core_web_sm
```

```
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.11)
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.15)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.13)
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Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (8.3.10)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.1.3)
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.5.2)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.10)
Requirement already satisfied: weasel<0.5.0,>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.4.3)
Requirement already satisfied: typer-slim<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.24.0)
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (4.67.3)
Requirement already satisfied: numpy>=1.19.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.2)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.32.4)
Requirement already satisfied: pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.1.6)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from spacy) (75.2.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (26.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.24.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets) (3.6.0)
Requirement already satisfied: multiprocess<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>
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Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spac
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (
Requirement already satisfied: bliss<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->sp
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Requirement already satisfied: typer>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from typer-slim<1.0.0,>=0.3.0->sp
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=
```

STEP 3 — Import Libraries

```
import spacy                    # NLP library for tokenization & NER
from spacy import displacy      # For visualization of entities

from transformers import pipeline, AutoTokenizer, AutoModelForTokenClassification
import torch                    # Deep learning framework
import pandas as pd             # Data handling
```

STEP 4 — Download and Load spaCy Model

```
nlp = spacy.load("en_core_web_sm")
```

STEP 5 — Prepare Sample Text Data

```
sentences = [
    "Microsoft announced a new AI product in California.",
    "Virat Kohli scored a century in Mumbai.",
    "Prime Minister Narendra Modi visited the United States.",
    "Google launched Gemini AI in New York.",
    "The World Health Organization held a meeting in Geneva."
]
```

STEP 6 — Apply spaCy NER

```
for sentence in sentences:
    doc = nlp(sentence)
    print("\nSentence:", sentence)
    for ent in doc.ents:
        print("Entity:", ent.text, "| Label:", ent.label_)
```

```
Sentence: Microsoft announced a new AI product in California.
Entity: Microsoft | Label: ORG
Entity: AI | Label: ORG
Entity: California | Label: GPE

Sentence: Virat Kohli scored a century in Mumbai.
Entity: Virat Kohli | Label: PERSON
Entity: a century | Label: DATE
Entity: Mumbai | Label: GPE

Sentence: Prime Minister Narendra Modi visited the United States.
Entity: Narendra Modi | Label: PERSON
Entity: the United States | Label: GPE

Sentence: Google launched Gemini AI in New York.
Entity: Google | Label: ORG
Entity: Gemini AI | Label: ORG
Entity: New York | Label: GPE

Sentence: The World Health Organization held a meeting in Geneva.
Entity: The World Health Organization | Label: ORG
Entity: Geneva | Label: GPE
```

STEP 7 — Store spaCy Results in DataFrame

```
data = []

for sentence in sentences:
    doc = nlp(sentence)
    for ent in doc.ents:
        data.append([sentence, ent.text, ent.label_])

df_spacy = pd.DataFrame(data, columns=["Sentence", "Entity", "Label"])
df_spacy
```

	Sentence	Entity	Label
0	Microsoft announced a new AI product in Califo...	Microsoft	ORG
1	Microsoft announced a new AI product in Califo...	AI	ORG
2	Microsoft announced a new AI product in Califo...	California	GPE
3	Virat Kohli scored a century in Mumbai.	Virat Kohli	PERSON
4	Virat Kohli scored a century in Mumbai.	a century	DATE
5	Virat Kohli scored a century in Mumbai.	Mumbai	GPE
6	Prime Minister Narendra Modi visited the Unite...	Narendra Modi	PERSON
7	Prime Minister Narendra Modi visited the Unite...	the United States	GPE
8	Google launched Gemini AI in New York.	Google	ORG
9	Google launched Gemini AI in New York.	Gemini AI	ORG
10	Google launched Gemini AI in New York.	New York	GPE
11	The World Health Organization held a meeting i...	The World Health Organization	ORG
12	The World Health Organization held a meeting i...	Geneva	GPE

STEP 8 — (Optional) Visualize Entities

```
doc = nlp(sentences[0])
displacy.render(doc, style="ent", jupyter=True)
```

Microsoft ORG announced a new AI ORG product in California GPE .

STEP 9 — Load Hugging Face NER Model

```
model_name = "dbmdz/bert-large-cased-finetuned-conll03-english"

tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForTokenClassification.from_pretrained(model_name)

ner_pipeline = pipeline("ner", model=model, tokenizer=tokenizer)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
config.json: 100%  998/998 [00:00<00:00, 9.32kB/s]
tokenizer_config.json: 100%  60.0/60.0 [00:00<00:00, 982B/s]
vocab.txt: 213k/? [00:00<00:00, 1.49MB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
model.safetensors: 100%  1.33G/1.33G [00:21<00:00, 242MB/s]
Loading weights: 100%  391/391 [00:01<00:00, 399.31it/s, Materializing param=classifier.weight]
BertForTokenClassification LOAD REPORT from: dbmdz/bert-large-cased-finetuned-conll03-english
Key                       | Status     |  |
-------------------------+------------+--+-
bert.pooler.dense.bias    | UNEXPECTED |  |
bert.pooler.dense.weight  | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

STEP 10 — Apply Transformer NER

```
for sentence in sentences:
    print("\nSentence:", sentence)
    results = ner_pipeline(sentence)
    for r in results:
        print("Word:", r['word'],
              "| Label:", r['entity'],
              "| Score:", round(r['score'],4))
```

```
Sentence: Microsoft announced a new AI product in California.
Word: Microsoft | Label: I-ORG | Score: 0.9994
Word: AI | Label: I-ORG | Score: 0.8433
Word: California | Label: I-LOC | Score: 0.9998

Sentence: Virat Kohli scored a century in Mumbai.
Word: V | Label: I-PER | Score: 0.9996
Word: ##ira | Label: I-PER | Score: 0.9958
Word: ##t | Label: I-PER | Score: 0.999
Word: Ko | Label: I-PER | Score: 0.9995
Word: ##hl | Label: I-PER | Score: 0.9913
Word: ##i | Label: I-PER | Score: 0.9952
Word: Mumbai | Label: I-LOC | Score: 0.9977

Sentence: Prime Minister Narendra Modi visited the United States.
Word: Na | Label: I-PER | Score: 0.9992
Word: ##ren | Label: I-PER | Score: 0.9921
Word: ##dra | Label: I-PER | Score: 0.9983
Word: Mo | Label: I-PER | Score: 0.9989
Word: ##di | Label: I-PER | Score: 0.9808
Word: United | Label: I-LOC | Score: 0.9997
Word: States | Label: I-LOC | Score: 0.9998

Sentence: Google launched Gemini AI in New York.
Word: Google | Label: I-ORG | Score: 0.999
Word: Gemini | Label: I-ORG | Score: 0.9944
Word: AI | Label: I-ORG | Score: 0.9916
Word: New | Label: I-LOC | Score: 0.9993
Word: York | Label: I-LOC | Score: 0.9992

Sentence: The World Health Organization held a meeting in Geneva.
Word: World | Label: I-ORG | Score: 0.9996
```

```
Word: Health | Label: I-ORG | Score: 0.9993
Word: Organization | Label: I-ORG | Score: 0.9993
Word: Geneva | Label: I-LOC | Score: 0.9994
```

STEP 11 — Clean Transformer Output

```
clean_data = []

for sentence in sentences:
    results = ner_pipeline(sentence)
    for r in results:
        word = r['word'].replace("##", "")
        clean_data.append([
            sentence,
            word,
            r['entity'],
            round(r['score'],4)
        ])

df_hf = pd.DataFrame(clean_data,
                     columns=["Sentence", "Entity", "Label", "Confidence"])
df_hf
```

	Sentence	Entity	Label	Confidence
0	Microsoft announced a new AI product in Califo...	Microsoft	I-ORG	0.9994
1	Microsoft announced a new AI product in Califo...	AI	I-ORG	0.8433
2	Microsoft announced a new AI product in Califo...	California	I-LOC	0.9998
3	Virat Kohli scored a century in Mumbai.	V	I-PER	0.9996
4	Virat Kohli scored a century in Mumbai.	ira	I-PER	0.9958
5	Virat Kohli scored a century in Mumbai.	t	I-PER	0.9990
6	Virat Kohli scored a century in Mumbai.	Ko	I-PER	0.9995
7	Virat Kohli scored a century in Mumbai.	hl	I-PER	0.9913
8	Virat Kohli scored a century in Mumbai.	i	I-PER	0.9952
9	Virat Kohli scored a century in Mumbai.	Mumbai	I-LOC	0.9977
10	Prime Minister Narendra Modi visited the Unite...	Na	I-PER	0.9992
11	Prime Minister Narendra Modi visited the Unite...	ren	I-PER	0.9921
12	Prime Minister Narendra Modi visited the Unite...	dra	I-PER	0.9983
13	Prime Minister Narendra Modi visited the Unite...	Mo	I-PER	0.9989
14	Prime Minister Narendra Modi visited the Unite...	di	I-PER	0.9808
15	Prime Minister Narendra Modi visited the Unite...	United	I-LOC	0.9997
16	Prime Minister Narendra Modi visited the Unite...	States	I-LOC	0.9998
17	Google launched Gemini AI in New York.	Google	I-ORG	0.9990
18	Google launched Gemini AI in New York.	Gemini	I-ORG	0.9944
19	Google launched Gemini AI in New York.	AI	I-ORG	0.9916
20	Google launched Gemini AI in New York.	New	I-LOC	0.9993
21	Google launched Gemini AI in New York.	York	I-LOC	0.9992
22	The World Health Organization held a meeting i...	World	I-ORG	0.9996
23	The World Health Organization held a meeting i...	Health	I-ORG	0.9993
24	The World Health Organization held a meeting i...	Organization	I-ORG	0.9993
25	The World Health Organization held a meeting i...	Geneva	I-LOC	0.9994

STEP 12 — Create Comparison Table

Feature	spaCy	Hugging Face
Model Type	Statistical + Rule-based	Transformer (BERT)
Speed	Fast	Slower
Accuracy	Good	Higher
Context Handling	Moderate	Strong
Confidence Score	No	Yes
GPU Requirement	No	Recommended

STEP 13 — Result Analysis

Example Points:

Transformer model detected more detailed entities.

spaCy is faster and lightweight.

Transformer provides probability scores.

BERT handles context better due to attention mechanism.

spaCy suitable for real-time applications.

Transformers require more memory and GPU.

Hugging Face model showed better generalization.

Training time is higher for Transformers.

STEP 14 — Error Analysis

Example:

❌ Error 1

Sentence: "Virat Kohli scored a century in Mumbai." spaCy missed "century" context as event. Reason: Model trained mainly for PERSON, ORG, GPE types.

❌ Error 2

Transformer split “Narendra Modi” into subwords. Reason: WordPiece tokenization.

❌ Error 3

"Gemini AI" sometimes detected partially. Reason: New/rare entity not in training dataset.